

Math Camp

Lecture 4

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Prices as linear functionals

- A commodity bundle is a vector $\mathbf{x} \in \mathbb{R}_+^n$: x_i units of good i .
- A price system assigns to each bundle its cost,

$$p(\mathbf{x}) = \sum_{i=1}^n p_i x_i = \mathbf{p}' \mathbf{x}$$

- This map is linear.
 - $p(\mathbf{x} + \mathbf{y}) = p(\mathbf{x}) + p(\mathbf{y})$: buying two bundles costs the sum of their costs.
 - $p(\alpha \mathbf{x}) = \alpha p(\mathbf{x})$: doubling the bundle doubles the bill.

Prices as linear functionals

- So $p \in (\mathbb{R}^n)^*$: a price is an element of the *dual*, not of the commodity space.
 - $p_i = p(\mathbf{e}_i)$: the coefficients are the cost of one unit of good i , recovered by evaluating p on the canonical basis.
 - $\mathbf{p}'\mathbf{x}$ is a *pairing* of two different spaces, not an inner product of like objects.

Column and Row Space

Let $M \in \mathbb{R}^{m \times n}$, and denote by $\{M^i\}_{i=1}^m \subset \mathbb{R}^n$ the rows of M , and by $\{M_j\}_{j=1}^n \subset \mathbb{R}^m$ the columns of M .

- The *row space* of M is $\text{span}\left(\{M^i\}_{i=1}^m\right)$
- The *column space* of M is $\text{span}\left(\{M_j\}_{j=1}^n\right)$
- Let $T : \mathbf{V} \rightarrow \mathbf{W}$ be represented by M . Then $T(\mathbf{V})$ is the column space of M .
 - All of the information transferred by the linear map is expressed as linear combinations of the columns of M .

Column and Row Space

$$T : (v_1 \ v_2 \ v_3) \rightarrow \begin{pmatrix} v_1 + 2v_2 + 3v_3 \\ 2v_1 + 4v_2 + v_3 \end{pmatrix}$$

Column and Row Space

- The dimensions of the column and row spaces are equal. Take a dual perspective.
- $\mathbf{w}^* \in \ker(T^*)$ iff $T(\mathbf{V}) \subseteq \ker(\mathbf{w}^*)$, and $\dim(\ker(\mathbf{w}^*)) = m - 1$ for $\mathbf{w}^* \neq 0$.
- So there are $m - \dim(T(\mathbf{V}))$ linearly independent functionals in $\mathbf{W}^*$ satisfying $T(\mathbf{V}) \subseteq \ker(\mathbf{w}^*)$, whence $\dim(\ker(T^*)) = m - \dim(T(\mathbf{V}))$ and, by rank-nullity,

$$\dim(T^*(\mathbf{W}^*)) = \dim(T(\mathbf{V}))$$

Column and Row Space

$$\begin{aligned}\dim \text{ of row space of } M &= \dim \text{ of column space of } M' \\ &= \dim(T^*(\mathbf{W}^*)) \\ &= \dim(T(\mathbf{V})) \\ &= \dim \text{ of column space of } M\end{aligned}$$

- This common number is the *rank*.

Coordinate Transforms

- Let $\mathbf{V}$ be a vector space with bases $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ and $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$
 - For $\mathbf{v} \in \mathbf{V}$, α^v and β^v are the coordinate representations of $\mathbf{v}$ against A and B .
 - $\beta(A, B)$ is the $n \times n$ matrix whose columns are the coordinate representations of the elements of A against the basis B : $\mathbf{a}_j = \sum_i \beta(A, B)_{ij} \mathbf{b}_i$

$$\mathbf{v} = \sum_j \alpha_j^v \mathbf{a}_j = \sum_j \alpha_j^v \left(\sum_i \beta(A, B)_{ij} \mathbf{b}_i \right) = \sum_i \left(\sum_j \beta(A, B)_{ij} \alpha_j^v \right) \mathbf{b}_i$$

- So $\beta^v = \beta(A, B) \alpha^v$

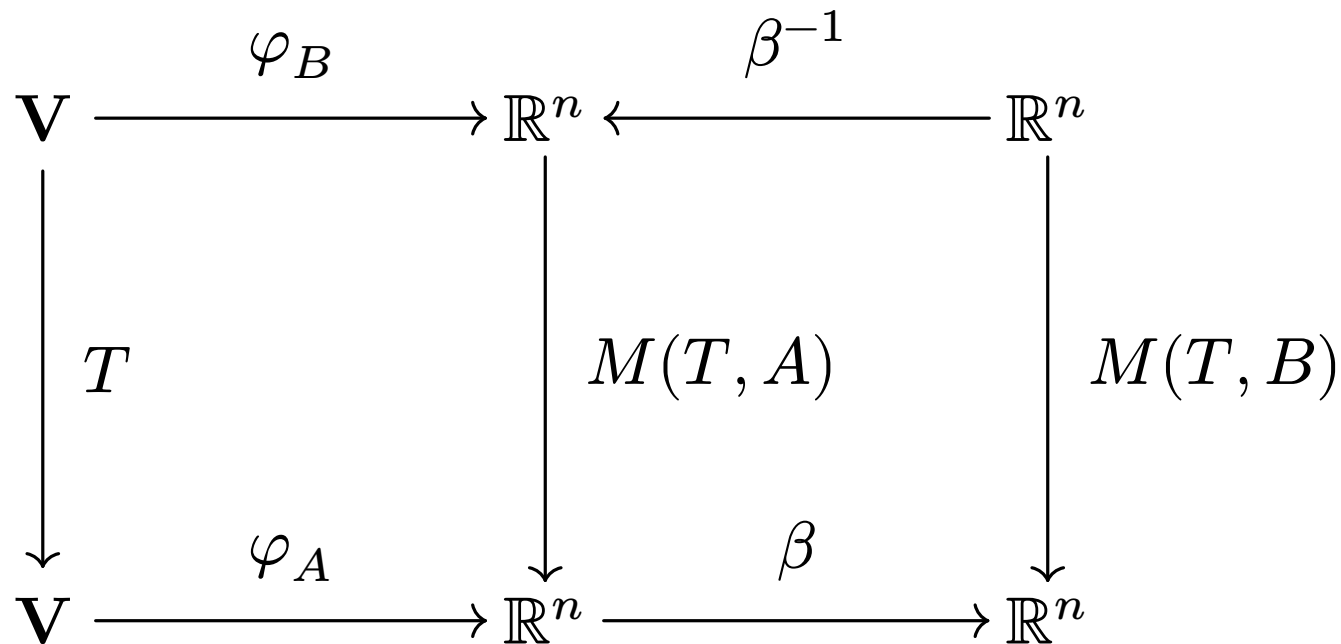
Coordinate Transforms

Let $A = \{(4, 1), (1, 4)\}$, $B = \{(1, 1), (2, -1)\}$ and $\alpha^v = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ be the co-ordinate representation of $\mathbf{v}$ on A

Coordinate Transforms

- Let $T : \mathbf{V} \rightarrow \mathbf{V}$ be linear, and let $M(T, A)$ and $M(T, B)$ be its matrix representations against A and B . Then

$$M(T, B) = \beta(A, B)M(T, A)\beta(A, B)^{-1}$$



Coordinate Transforms

- Two matrices P and Q are *similar* if there is an invertible S so that $Q = SPS^{-1}$
 - Two matrices are similar provided they represent the same linear transform against two (possibly) different bases.
 - Similar matrices share determinant, rank, characteristic polynomial and eigenvalues: these belong to the map, not the basis.

Coordinate Transforms

$$T : \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 5x-2y \\ 3x \end{pmatrix}$$

Coordinate Transforms

- Let M be an $n \times n$ matrix representing $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ against the canonical basis.
 - Suppose M has n linearly independent eigenvectors $\Xi = \{\xi_1, \dots, \xi_n\}$ and associated eigenvalues $\{\lambda_1, \dots, \lambda_n\}$.
 - Let S be the matrix whose columns are the ξ_i . Then S changes coordinates from Ξ to the canonical basis.
 - Let Λ be the matrix representation of T against Ξ .
 - $M = S\Lambda S^{-1}$, so $\Lambda = S^{-1}MS$.
 - MS has columns $\lambda_i \xi_i$, so $\Lambda_{ij} = \delta_{ij} \lambda_i$: Λ is diagonal with the eigenvalues on the diagonal.

Coordinate Transforms

- The product $S\Lambda S^{-1}$ is the *eigenvalue decomposition* of M .
 - The eigenspaces are the natural basis against which to measure a linear map: on them, the map simply scales.
 - $M^k = S\Lambda^k S^{-1}$, and Λ^k is diagonal with entries λ_i^k : powers of a matrix become powers of scalars.
- When an eigenvalue decomposition exists the matrix is *diagonalizable*, i.e. similar to a diagonal matrix.
- Not all matrices are diagonalizable. Suppose $M \neq 0$ and $M^m = 0$ for some $m \in \mathbb{N}$.
 - If $M = S\Lambda S^{-1}$ then $\Lambda \neq 0$, so $\Lambda_{ii} = \lambda_i \neq 0$ for some i .
 - $M^m = S\Lambda^m S^{-1}$ and $(\Lambda^m)_{ij} = \delta_{ij}\lambda_i^m$, so $(\Lambda^m)_{ii} \neq 0$, so $M^m \neq 0$. Contradiction.

Coordinate Transforms

- **Theorem**

If $M \in \mathbb{R}^{n \times n}$ has n distinct eigenvalues then M is diagonalizable.

- Matrices are generically diagonalizable.
- Nobody said the entries of Λ are real.
- Computers compute eigenvalues and eigenvectors very efficiently.

Matrix structure

- M is *upper triangular* if $M_{ij} = 0$ whenever $i > j$.
- A complex M is *unitary* if $M^* M = I_n = M M^*$, where M^* is the Hermitian transpose. A real M is *orthogonal* if $M' M = I_n = M M'$.
 - For a unitary or orthogonal matrix, M^{-1} is just the transpose.
- M is *block diagonal* if it can be written as a diagonal matrix of square matrices,

$$M = \begin{pmatrix} M_1 & 0 & \dots & 0 \\ 0 & M_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & M_k \end{pmatrix} \equiv M_1 \oplus \dots \oplus M_k$$

with all 0s representing conformable matrices of zeros.

Matrix structure

- Let $T : \mathbf{V} \rightarrow \mathbf{V}$ and $S : \mathbf{W} \rightarrow \mathbf{W}$ be linear. The *direct sum* is

$$(T \oplus S)(\mathbf{v}, \mathbf{w}) = (T(\mathbf{v}), S(\mathbf{w}))$$

- ▶ If A and B are bases for $\mathbf{V}$ and $\mathbf{W}$, then $A \oplus B = \{(\mathbf{a}_1, 0), \dots, (\mathbf{a}_n, 0), (0, \mathbf{b}_1), \dots, (0, \mathbf{b}_m)\}$ is a basis for $\mathbf{V} \oplus \mathbf{W}$.
- ▶ If P and Q represent T and S against A and B , then $P \oplus Q$ represents $T \oplus S$ against $A \oplus B$.
- $\mathbf{V} \oplus 0 = \{(\mathbf{v}, 0) : \mathbf{v} \in \mathbf{V}\}$ is a subspace with $(T \oplus S)(\mathbf{V} \oplus 0) \subseteq \mathbf{V} \oplus 0$: an *invariant subspace*.

Matrix structure

- If an $n \times n$ matrix P is similar to a block diagonal M then there are k subspaces of $\mathbb{R}^n$ invariant under M .
 - After changing coordinates, the action of P on these subspaces is decoupled, so each can be examined in isolation.
 - To understand P we only need understand M ; to understand M , we only need the M_i .

Matrix structure

$$M = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Matrix structure

- An upper triangular matrix gives a *nested* sequence of subspaces rather than a decoupling.
- Let $M \in \mathbb{R}^{n \times n}$ be upper triangular and let $\mathbf{V}_i \subset \mathbb{R}^n$ be spanned by $\{\mathbf{e}_1, \dots, \mathbf{e}_i\}$. Then $\mathbf{v} \in \mathbf{V}_i \Rightarrow M\mathbf{v} \in \mathbf{V}_i$.
 - To understand M we only need its action on each $\mathbf{V}_i$.
 - Given the action on $\mathbf{V}_{i-1}$, determining the action on $\mathbf{V}_i$ requires understanding only one new dimension.

Matrix structure

$$M = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Decompositions

- **Schur's decomposition:**

Let V be finite dimensional and $f : V \rightarrow V$ be linear. Then there is a collection of invariant subspaces $\{V_k\}_{k=1}^n$ such that

$$\dim(V_k) = k \text{ and } V_k \subset V_{k+1}$$

- ▶ We can study f recursively, and each step in the recursion adds only one dimension, i.e. one new basis element on which f must be examined.

Decompositions

- For λ in $\mathbb{C}$ or $\mathbb{R}$ and $n \in \mathbb{N}$, the *Jordan block* is

$$J(\lambda, n) = \begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

- If $\lambda = a + bi$ with $b \neq 0$, define

$$C(\lambda) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

- $C(\lambda)$ is a rotation together with a scaling

Decompositions

- **Theorem**

Let $M \in \mathbb{R}^{n \times n}$ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$.

- M is similar to a Jordan block diagonal matrix,

$$J(\lambda_{m_1}, n_1) \oplus \dots \oplus J(\lambda_{m_N}, n_N) = \begin{pmatrix} J(\lambda_{m_1}, n_1) & & & \\ & J(\lambda_{m_2}, n_2) & & \\ & & \ddots & \\ & & & J(\lambda_{m_N}, n_N) \end{pmatrix}$$

.

- ▶ There may be multiple Jordan blocks associated to the same eigenvalue, hence the m_k subscript.

Decompositions

- ▶ If λ is not real and $J(\lambda, m)$ is an associated Jordan block then there is necessarily a block $J(\bar{\lambda}, m)$. The direct sum $J(\lambda, m) \oplus J(\bar{\lambda}, m)$ can be replaced by a $2m$ block

$$J^{\mathbb{R}}(\lambda, 2m) = \begin{pmatrix} C(\lambda) & I_2 & & \\ & C(\lambda) & \ddots & \\ & & \ddots & I_2 \\ & & & C(\lambda) \end{pmatrix}$$

Decompositions

- The Jordan decomposition provides a change of coordinates that decomposes the vector space into the smallest possible invariant subspaces, and the matrix into a block diagonal acting on each in isolation.
 - The eigenvalue decomposition is the special case where every block is 1×1 .
- Fully general, but for computational purposes a Schur decomposition is often preferable.
 - The Jordan form is discontinuous in the entries: an arbitrarily small perturbation can change it entirely.

Decompositions

- **Theorem**

Let $M \in \mathbb{R}^{n \times n}$ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$.

- ▶ $M = QTQ^*$ where T is upper triangular with the eigenvalues of M on the diagonal and Q is unitary.
- ▶ $M = ZT^{\mathbb{R}}Z'$ where $T^{\mathbb{R}}$ is upper block triangular and Z is orthogonal. Real eigenvalues correspond to 1×1 blocks; conjugate pairs $(\lambda, \bar{\lambda})$ correspond to 2×2 blocks $C(\lambda)$.
- A Schur decomposition gives a nested sequence of invariant subspaces together with a representation acting on them recursively.
- Matrix inversion, which is computationally expensive, is not needed — just take the transpose.

Definiteness

- Fix $M \in \mathbb{R}^{n \times n}$. We may view M as a map from $\mathbb{R}^n \oplus \mathbb{R}^n$ to $\mathbb{R}$ by sending $(\mathbf{v}, \mathbf{w})$ to $\mathbf{w}' M \mathbf{v}$.
 - For fixed $\mathbf{w}$, the map $\mathbf{v} \mapsto \mathbf{w}' M \mathbf{v}$ is a linear functional; for fixed $\mathbf{v}$, so is $\mathbf{w} \mapsto \mathbf{w}' M \mathbf{v}$.
 - Acting this way, M is a *bilinear form*.
- A linear form is a list with one index, $\mathbf{v} = (v_i)$; a bilinear form is a list with two, $M = (M_{ij})$.
 - A list with a multi-index is a *tensor*. A vector is a first-order tensor, a matrix a second-order tensor.
 - The k -th derivative of a real valued function on $\mathbb{R}^n$ evaluated at a point is a k -th order tensor.

Definiteness

- M is *symmetric* if $M = M'$.
- M is *positive definite* if $\mathbf{v}' M \mathbf{v} > 0$ for all non-zero $\mathbf{v} \in \mathbb{R}^n$.
- M is *positive semi-definite* if $\mathbf{v}' M \mathbf{v} \geq 0$ for all non-zero $\mathbf{v} \in \mathbb{R}^n$.
 - Analogous definitions hold for negative definiteness.

- **Theorem**

If M is symmetric and positive semi-definite then the eigenvalues of M are real and non-negative.

Definiteness

- An *inner product* on a real vector space V is a symmetric, positive definite, bilinear form.
 - On $\mathbb{R}^n$, every symmetric bilinear form is $\mathbf{v} \mapsto \mathbf{w}' M \mathbf{v}$ for a symmetric M .
 - So inner products on $\mathbb{R}^n$ and symmetric positive definite matrices are the same thing; the usual one is $M = I$.

Definiteness

- Positive and negative definiteness reflect concavity.
- Let $\mathbf{x} = (x_1, \dots, x_n)$ and let p be a quadratic function,

$$p(\mathbf{x}) = \sum_{ij} M_{ij} x_i x_j = \mathbf{x}' M \mathbf{x}$$

- ▶ If M is positive definite then $\mathbf{x} = 0$ is the unique minimum.
- The functions we minimize are not generally quadratic. But the second derivative of a real function on $\mathbb{R}^n$ evaluated at a point is a second-order tensor: a matrix, necessarily symmetric.
 - ▶ If this matrix is positive definite at a critical point, that point is a strict local minimum.